How to Think and Analyze Requirements

Understanding the problem before writing the solution

1. Read the requirement

Identify the main goal

Find inputs and expected outputs

Notice rules, conditions, and constraints



2. Break it into parts

1. Underline the important details
2. Separate what is given and what is needed
3. List the steps to solve it
4. Think of examples and edge cases

Analysis Notes

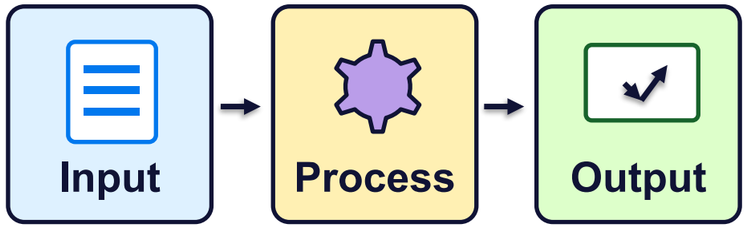
- ✅ Given: ...
- ✅ Needed: ...
- ✅ Steps: 1 → 2 → 3 ...
- ✅ Example / Edge case: ...

3. Check your understanding

Restate the requirement in simple words

Predict how the program should behave

Make sure nothing is missing



Key Idea

Good requirement analysis helps you understand the goal, the data, and the rules before coding.

Drawing a Flowchart

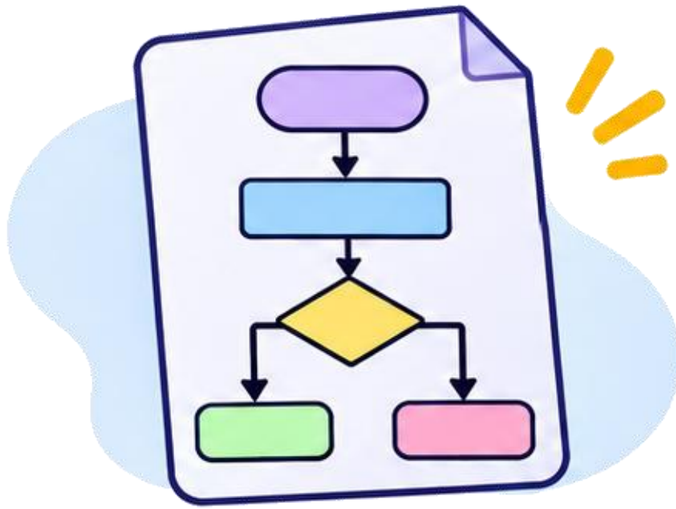
Visualizing the logic of a program step by step

1. What is a flowchart?

A flowchart shows the order of actions

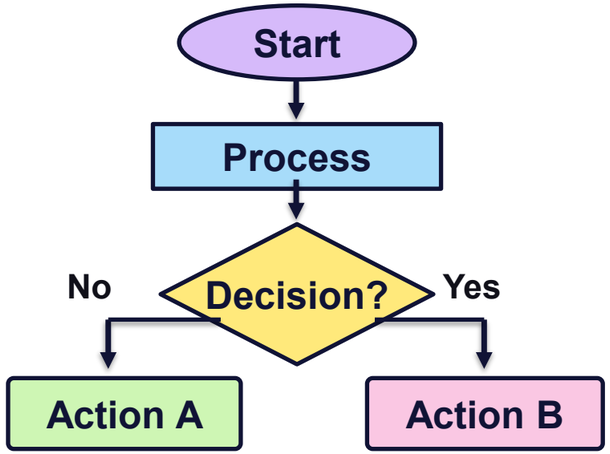
It uses symbols and arrows

It helps you see the logic clearly

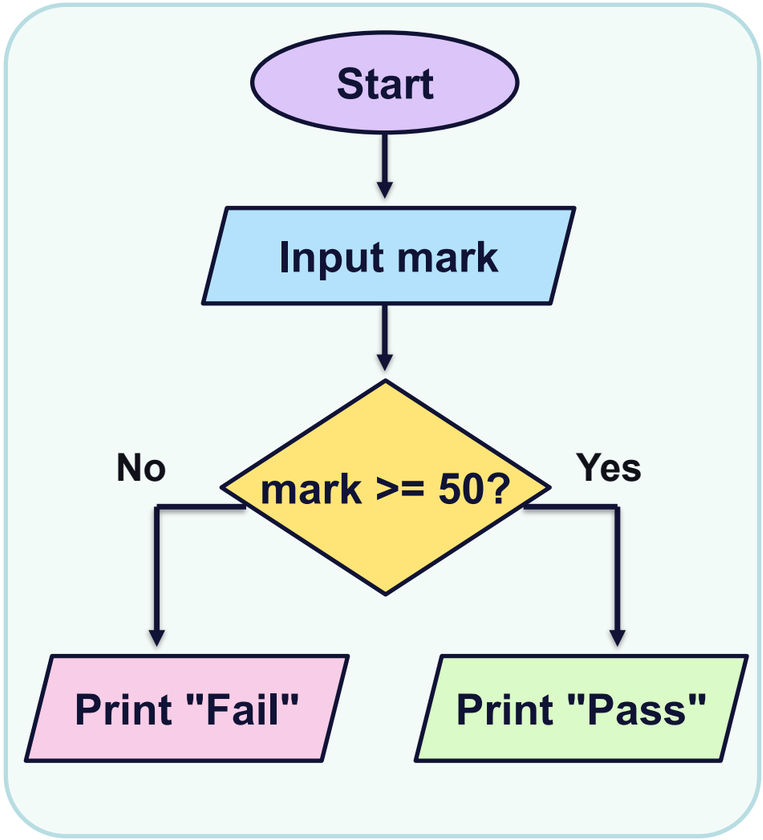


2. Common symbols

1. Start / End = Oval
2. Process = Rectangle
3. Decision = Diamond
4. Arrow = Direction of flow



3. Example



Key Idea | A flowchart turns program logic into an easy visual path.

Writing Pseudocode

Planning the logic before writing real code

1. What is pseudocode?

Pseudocode describes logic in simple words

It is not tied to one programming language

It helps you plan before coding



2. How to write it

1. Start with the problem goal
2. Write actions in order
3. Use IF, ELSE, and LOOP clearly
4. Keep it simple and readable

Pseudocode

```
START
INPUT mark
IF mark >= 50 THEN
   PRINT 'Pass'
ELSE
   PRINT 'Fail'
END IF
```

3. Why it helps

Makes thinking more organized

Helps you check the logic early

Makes coding easier later



Key Idea | **Pseudocode helps you think first, then code with confidence.**